

1. ArithmeticException

Question

Write a Java program to handle an **ArithmeticException** when dividing a number by zero.

Steps

1. Declare two numbers.
2. Divide the numbers inside **try**.
3. Handle division by zero using **catch**.
4. Display the error message.

Algorithm

1. Start.
2. Declare **a = 10** and **b = 0**.
3. Perform **a / b** inside **try**.
4. Catch **ArithmeticException**.
5. Display the error message.
6. Stop.

Program

```
public class ArithmeticExceptionDemo {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 0;  
  
        try {  
            int result = a / b;  
            System.out.println(result);  
        }  
    }  
}
```

```
    }  
    catch (ArithmeticException e) {  
        System.out.println("Cannot divide by zero");  
    }  
}  
}
```

Output

Cannot divide by zero

```
PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\477a7874c5187bcdd807e21f504eebd1\resources\bin\java.exe' -jar 'C:\Users\venik\OneDrive\Observation\task3\task3.jar'  
Cannot divide by zero  
PS C:\Users\venik\OneDrive\Observation\task3>
```

2. ArrayIndexOutOfBoundsException

Question

Write a Java program to handle **ArrayIndexOutOfBoundsException** when accessing an invalid array index.

Steps

1. Create an array.
2. Access an invalid index inside **try**.
3. Handle the exception using **catch**.
4. Display the error message.

Algorithm

1. Start.
2. Create an integer array.
3. Try to access an invalid index.
4. Catch **ArrayIndexOutOfBoundsException**.
5. Display the error message.
6. Stop.

Program

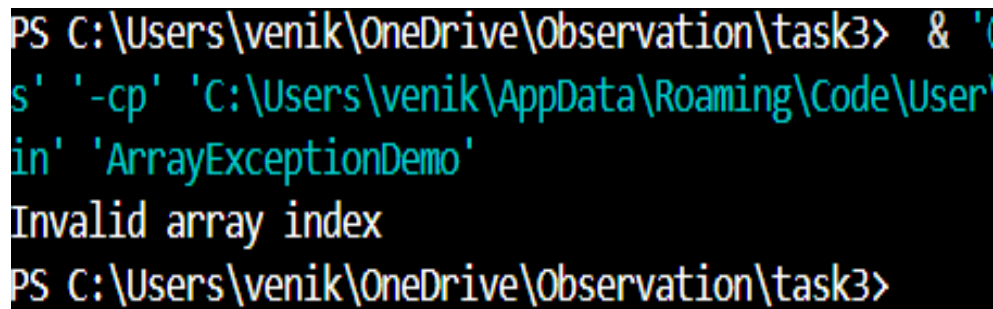
```
public class ArrayExceptionDemo {  
    public static void main(String[] args) {
```

```
int a[] = {10, 20, 30};

try {
    System.out.println(a[5]);
}
catch (ArrayIndexOutOfBoundsException e) {
    System.out.println("Invalid array index");
}
}
```

Output

Invalid array index



```
PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Users\venik\AppData\Roaming\Code\User\bin\ArrayExceptionDemo'
Invalid array index
PS C:\Users\venik\OneDrive\Observation\task3>
```

3. NumberFormatException

Question

Write a Java program to handle **NumberFormatException** when converting a string into an integer.

Steps

1. Create a string containing non-numeric data.
2. Convert it into an integer inside **try**.
3. Handle the exception using **catch**.
4. Display the error message.

Algorithm

1. Start.
2. Declare **str = "abc"**.
3. Convert the string using **Integer.parseInt()**.
4. Catch **NumberFormatException**.
5. Display the error message.
6. Stop.

Program

```
public class NumberFormatDemo {  
    public static void main(String[] args) {  
        String str = "abc";  
  
        try {
```

```
        int num = Integer.parseInt(str);
        System.out.println(num);
    }
    catch (NumberFormatException e) {
        System.out.println("Invalid number format");
    }
}
}
```

Output

Invalid number format

```
PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Program Files\Java\jdk-25.
s' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\477a7874c51
in' 'NumberFormatDemo'
Invalid number format
PS C:\Users\venik\OneDrive\Observation\task3>
```

4. NullPointerException

Question

Write a Java program to handle **NullPointerException** when accessing a null string.

Steps

1. Declare a string as **null**.
2. Try to access the string.
3. Handle the exception.
4. Display the error message.

Algorithm

1. Start.
2. Declare **str = null**.
3. Try to find its length.
4. Catch **NullPointerException**.
5. Display the error message.
6. Stop.

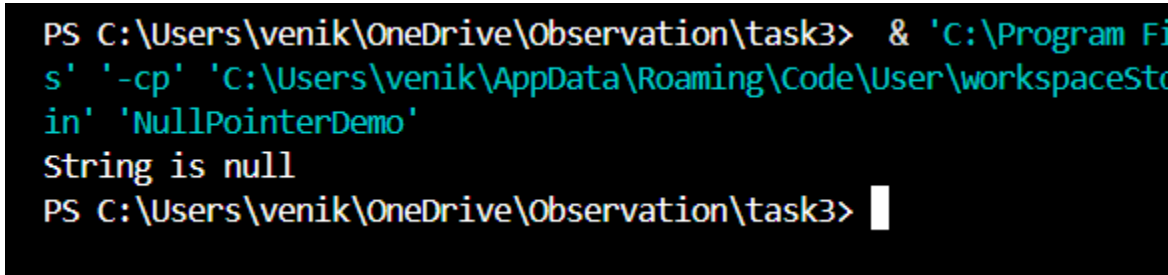
Program

```
public class NullPointerExceptionDemo {  
    public static void main(String[] args) {  
        String str = null;  
  
        try {
```

```
        System.out.println(str.length());
    }
    catch (NullPointerException e) {
        System.out.println("String is null");
    }
}
}
```

Output

String is null



```
PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Program Files'
s' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceSto
in' 'NullPointerExceptionDemo'
String is null
PS C:\Users\venik\OneDrive\Observation\task3> |
```

5. StringIndexOutOfBoundsException

Question

Write a Java program to handle **StringIndexOutOfBoundsException** when accessing an invalid string index.

Steps

1. Create a string.
2. Access an invalid character position.
3. Handle the exception.
4. Display the error message.

Algorithm

1. Start.
2. Declare `str = "Hello"`.
3. Try to access index `10`.
4. Catch **StringIndexOutOfBoundsException**.
5. Display the error message.
6. Stop.

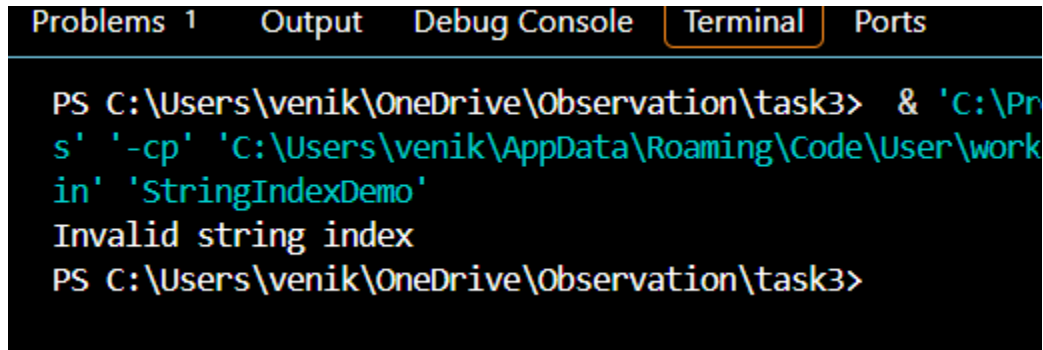
Program

```
public class StringIndexDemo {
```

```
public static void main(String[] args) {  
    String str = "Hello";  
  
    try {  
        System.out.println(str.charAt(10));  
    }  
    catch (StringIndexOutOfBoundsException e) {  
        System.out.println("Invalid string index");  
    }  
}
```

Output

Invalid string index



```
Problems 1 Output Debug Console Terminal Ports  
PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Pr  
s' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\work  
in' 'StringIndexDemo'  
Invalid string index  
PS C:\Users\venik\OneDrive\Observation\task3>
```

6. NegativeArraySizeException

Question

Write a Java program to handle **NegativeArraySizeException** when creating an array with a negative size.

Steps

1. Give a negative size for an array.
2. Create the array inside **try**.
3. Handle the exception.
4. Display the error message.

Algorithm

1. Start.
2. Declare **size = -5**.
3. Create an array using the size.
4. Catch **NegativeArraySizeException**.
5. Display the error message.
6. Stop.

Program

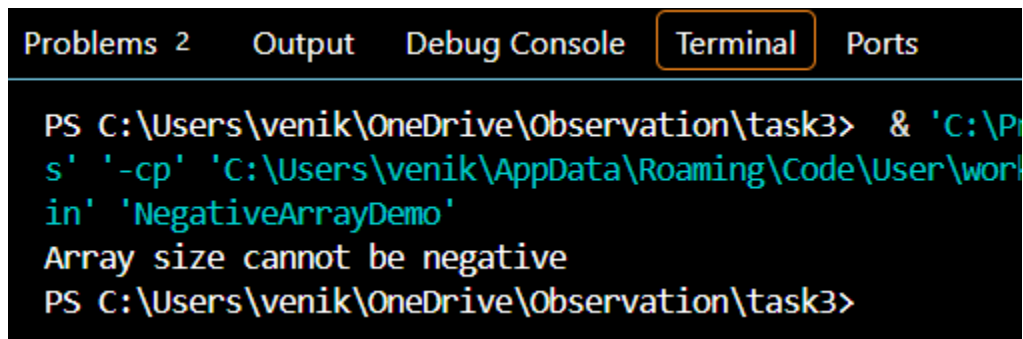
```
public class NegativeArrayDemo {  
    public static void main(String[] args) {
```

```
int size = -5;

try {
    int a[] = new int[size];
}
catch (NegativeArraySizeException e) {
    System.out.println("Array size cannot be negative");
}
}
```

Output

Array size cannot be negative



```
Problems 2  Output  Debug Console  Terminal  Ports

PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspace\NegativeArrayDemo'
Array size cannot be negative
PS C:\Users\venik\OneDrive\Observation\task3>
```

7. ClassCastException

Question

Write a Java program to handle **ClassCastException** when an object is incorrectly converted to another class type.

Steps

1. Create an object.
2. Try to convert it into an incorrect type.
3. Handle the exception.
4. Display the error message.

Algorithm

1. Start.
2. Create a **String** object.
3. Try to convert it into an **Integer**.
4. Catch **ClassCastException**.
5. Display the error message.
6. Stop.

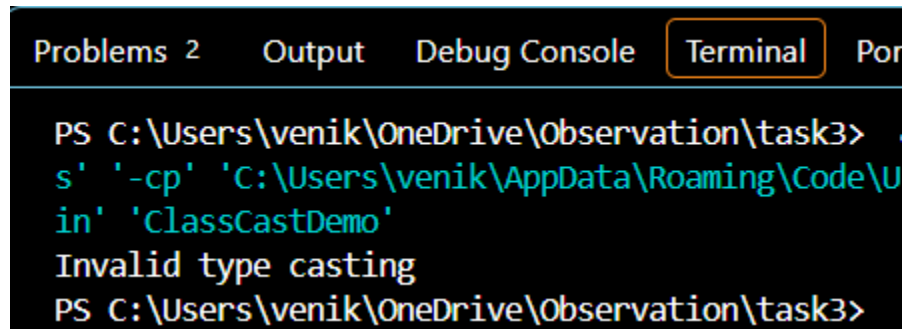
Program

```
public class ClassCastDemo {  
    public static void main(String[] args) {  
        Object obj = "Hello";  
  
        try {  
            Integer num = (Integer) obj;  
            System.out.println(num);  
        }  
        catch (ClassCastException e) {  
            System.out.println("Invalid type casting");  
        }  
    }  
}
```

```
}  
}  
}
```

Output

Invalid type casting



```
Problems 2  Output  Debug Console  Terminal  Port  
PS C:\Users\venik\OneDrive\Observation\task3> .\ClassCastDemo.exe  
s' '-cp' 'C:\Users\venik\AppData\Roaming\Code\U  
in' 'ClassCastDemo'  
Invalid type casting  
PS C:\Users\venik\OneDrive\Observation\task3>
```

8. InputMismatchException

Question

Write a Java program to handle **InputMismatchException** when the user enters an incorrect data type.

Steps

1. Create a **Scanner** object.
2. Ask the user to enter an integer.
3. Read the input inside **try**.
4. Handle incorrect input using **catch**.
5. Display the error message.

Algorithm

1. Start.
2. Create a Scanner object.
3. Ask the user to enter an integer.
4. Read the integer inside **try**.
5. Catch **InputMismatchException**.
6. Display the error message.
7. Stop.

Program

```

import java.util.Scanner;
import java.util.InputMismatchException;

public class InputMismatchDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter a number: ");
            int num = sc.nextInt();
            System.out.println("Number: " + num);
        }
        catch (InputMismatchException e) {
            System.out.println("Please enter an integer");
        }
    }
}

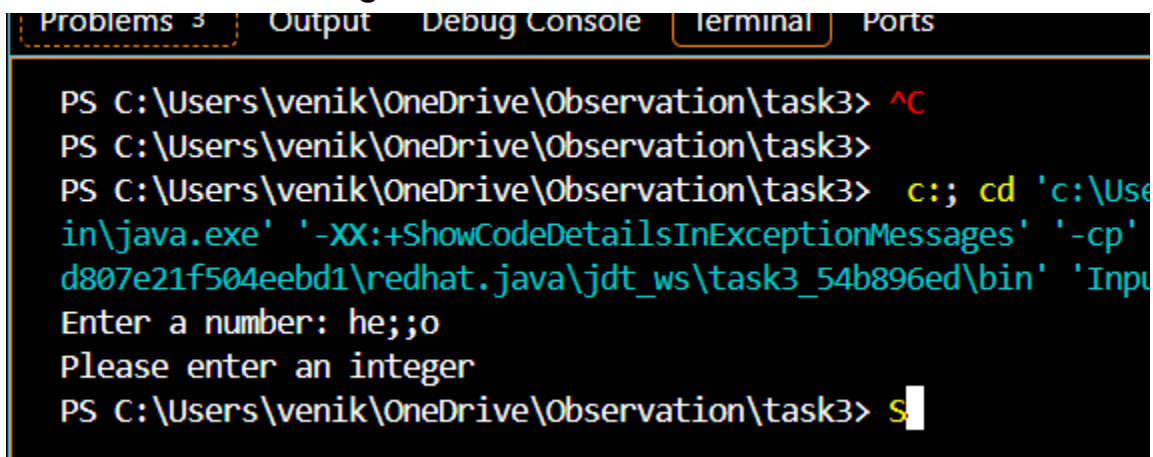
```

Output

If user enters **abc**:

Enter a number: abc

Please enter an integer



```

Problems 3 Output Debug Console Terminal Ports
PS C:\Users\venik\OneDrive\Observation\task3> ^C
PS C:\Users\venik\OneDrive\Observation\task3>
PS C:\Users\venik\OneDrive\Observation\task3> c:; cd 'c:\Use
in\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
d807e21f504eebd1\redhat.java\jdt_ws\task3_54b896ed\bin' 'Inpu
Enter a number: he;;o
Please enter an integer
PS C:\Users\venik\OneDrive\Observation\task3> S

```

9. IllegalArgumentException

Question

Write a Java program to handle **IllegalArgumentException** when an invalid argument is passed to a predefined method.

Steps

1. Create a string.
2. Use `substring()` with an invalid value.
3. Handle the exception.
4. Display the error message.

Algorithm

1. Start.
2. Declare a string.
3. Give an invalid starting position to `substring()`.
4. Catch **IllegalArgumentException**.
5. Display the error message.
6. Stop.

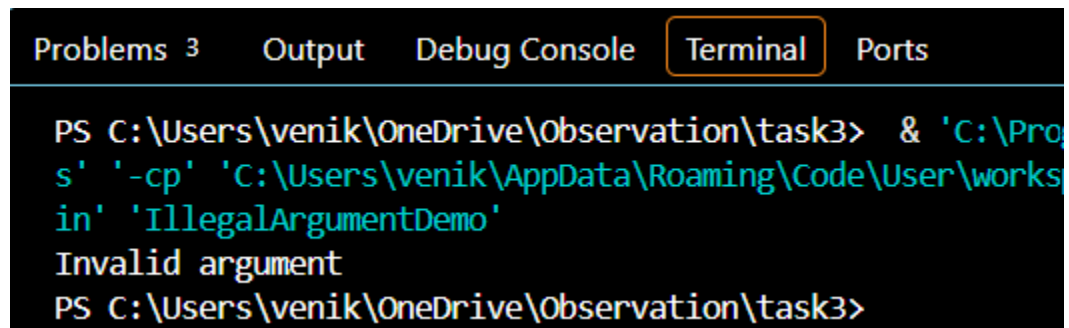
Program

```
public class IllegalArgumentExceptionDemo {  
    public static void main(String[] args) {  
        try {  
            Thread.sleep(-1000);  
        }  
        catch (IllegalArgumentException e) {  
            System.out.println("Invalid argument");  
        }  
    }  
}
```

```
        catch (InterruptedException e) {  
            System.out.println("Thread interrupted");  
        }  
    }  
}
```

Output

Invalid argument



The screenshot shows an IDE's interface with a dark theme. At the top, there is a tab bar with four tabs: 'Problems 3', 'Output', 'Debug Console', and 'Terminal'. The 'Terminal' tab is selected and highlighted with an orange border. Below the tab bar, the terminal window displays the following text:
PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' '-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspace\IllegalArgumentDemo'
Invalid argument
PS C:\Users\venik\OneDrive\Observation\task3>

10. Exception using `finally`

Question

Write a Java program to demonstrate the use of `try`, `catch`, and `finally` blocks.

Steps

1. Declare two numbers.
2. Perform division inside `try`.
3. Handle the exception using `catch`.
4. Execute the `finally` block.
5. Display the messages.

Algorithm

1. Start.
2. Declare `a = 10` and `b = 0`.
3. Perform division inside `try`.
4. Catch `ArithmeticException`.
5. Execute `finally`.
6. Stop.

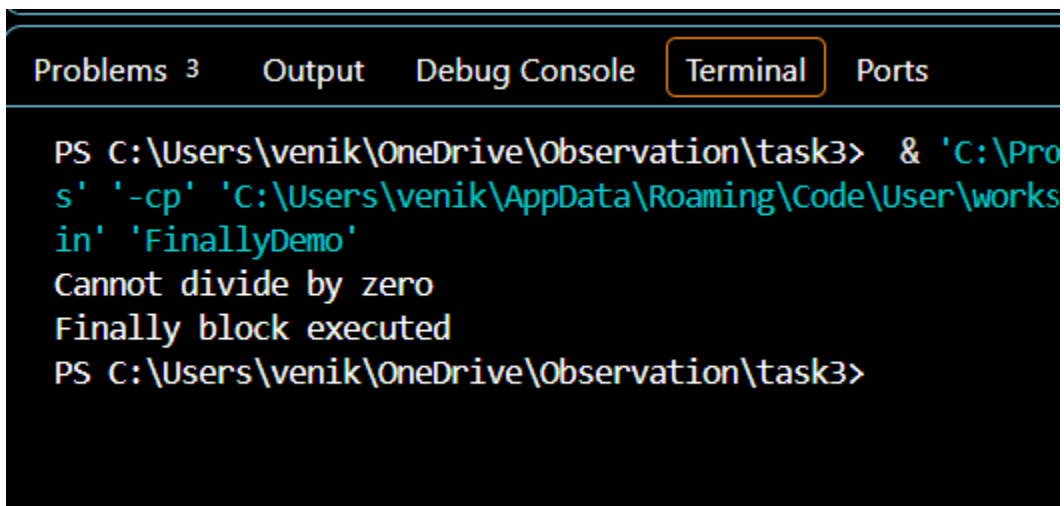
Program

```
public class FinallyDemo {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 0;  
  
        try {  
            int result = a / b;  
            System.out.println(result);  
        }  
        catch (ArithmeticException e) {  
            System.out.println("Cannot divide by zero");  
        }  
        finally {  
            System.out.println("Finally block executed");  
        }  
    }  
}
```

Output

Cannot divide by zero

Finally block executed



The screenshot shows a Visual Studio Code interface with a terminal window open. The terminal has tabs for 'Problems 3', 'Output', 'Debug Console', 'Terminal' (which is selected and highlighted with an orange border), and 'Ports'. The terminal output shows the command to run the program, followed by the runtime output: 'Cannot divide by zero' and 'Finally block executed', and then the command prompt again.

```
PS C:\Users\venik\OneDrive\Observation\task3> & 'C:\Program Files\Microsoft Visual Studio\2019\Community\MSBuild\Tools\dotnet.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspaceStorage\1\workspace\Task3\bin\Debug\netcoreapp3.1' 'FinallyDemo'  
Cannot divide by zero  
Finally block executed  
PS C:\Users\venik\OneDrive\Observation\task3>
```